

A Counterexample to Conjecture 00000000162: No Odd Polygon Has Two-Squares Sides for Primes $p \equiv 1 \pmod{4}$

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Abstract

Conjecture 00000000162 asserts that for every $n \geq 3$ there is a planar convex n -gon whose side vectors are two-squares vectors for primes $p \equiv 1 \pmod{4}$, i.e. integer vectors (a, b) with $a^2 + b^2 = p$, and sum to zero. This fails for every odd n , already for triangles. Such a vector has exactly one odd coordinate, so an odd number of them cannot sum to zero. The argument does not use convexity. It is formalized in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000000162 reads:

Conjecture: Convex polygons with prime-square side lengths: for every $n \geq 3$ there exists a planar convex n -gon whose side vectors all have prime squared lengths (two-squares vectors for primes $p \equiv 1 \pmod{4}$) and sum to zero.

2 Reading of the statement

The side vectors of a polygon with vertices $P_0, \dots, P_{n-1}$, in cyclic order, are $v_i = P_{i+1} - P_i$ (indices mod n). The phrase “prime squared lengths” is qualified in parentheses by “two-squares vectors for primes $p \equiv 1 \pmod{4}$ ”. We read this as specifying the side vectors, which is what the words say.

Definition 1. A *two-squares vector* for a prime $p \equiv 1 \pmod{4}$ is a vector $(a, b) \in \mathbb{Z}^2$ with $a^2 + b^2 = p$ for a prime $p \equiv 1 \pmod{4}$.

So the conjecture asserts: for every $n \geq 3$ there is a convex n -gon in the plane whose side vectors $v_0, \dots, v_{n-1}$ are two-squares vectors for primes $p \equiv 1 \pmod{4}$ and satisfy $\sum_i v_i = 0$. (The last condition holds for every closed polygon. It is part of the statement, and the refutation uses nothing else.)

The parenthetical is what gives the statement content. Without integrality it would be trivial: a regular n -gon with side $\sqrt{2}$ has all squared side lengths equal to the prime 2. Remark 5 discusses the reading that keeps integrality but also admits $p = 2$.

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3 Result

Theorem 2. *Let n be odd. There are no two-squares vectors $v_0, \dots, v_{n-1}$ for primes $p \equiv 1 \pmod{4}$ with $\sum_i v_i = 0$. In particular no n -gon, convex or not, has such side vectors, and Conjecture 00000000162 fails for $n = 3$.*

4 Proof

Lemma 3. *If $(a, b) \in \mathbb{Z}^2$ and $a^2 + b^2$ is odd, then $a + b$ is odd: exactly one of a, b is odd.*

Proof. $x^2 \equiv x \pmod{2}$ for every integer x , so $a + b \equiv a^2 + b^2 \equiv 1 \pmod{2}$. □

Proof of Theorem 2. Write $v_i = (a_i, b_i)$. Each $a_i^2 + b_i^2$ is a prime $\equiv 1 \pmod{4}$, hence odd, so $a_i + b_i$ is odd by Lemma 3. If $\sum_i v_i = 0$ then $\sum_i a_i = \sum_i b_i = 0$, so

$$0 = \sum_{i=0}^{n-1} (a_i + b_i) \equiv \sum_{i=0}^{n-1} 1 = n \pmod{2},$$

and n is even. □

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves its negation against Mathlib.

- `cross, IsConvexPolygon` — vertices $P: \text{Fin } n \rightarrow \mathbb{R}^2$ of a convex n -gon: $n \geq 3$, and for every edge $P_i \rightarrow P_{i+1}$ (successor `finRotate n`) all other vertices lie strictly on the same side, the same side for all edges.
- `IsTwoSquaresVector` — the definition above.
- `ConjectureHolds` — for every $n \geq 3$ there are P and $v: \text{Fin } n \rightarrow \mathbb{Z}^2$ with P convex, $P_{i+1} - P_i = v_i$, every v_i a two-squares vector for a prime $p \equiv 1 \pmod{4}$, and $\sum_i v_i = 0$.
- `odd_of_isTwoSquaresVector, odd_add_of_odd` — the squared length is odd, and Lemma 3 (computed in `ZMod 2`, where $x^2 = x$).
- `even_of_sum_eq_zero, not_exists_of_odd` — Theorem 2.
- `conjecture_00000000162_false` — $\neg \text{ConjectureHolds}$, at $n = 3$.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000000162_false
```

reports exactly `proptext, Classical.choice` and `Quot.sound`.

6 Remarks

Remark 4 (even n). For even $n = 2m$ the obstruction disappears, and the statement is plausibly true. Take m distinct primes $p_j \equiv 1 \pmod{4}$ and write $p_j = a_j^2 + b_j^2$. The vectors (a_j, b_j) are pairwise non-parallel, since distinct primes are not in a square ratio, and together with their negatives they are the side vectors of a centrally symmetric convex $2m$ -gon when sorted by angle. We do not formalize this.

Remark 5 (admitting $p = 2$). The obstruction is exactly the exclusion of the even prime. If $p = 2$ were admitted, so that $(\pm 1, \pm 1)$ could be used, odd polygons would exist. For example the triangle with vertices $(0, 0)$, $(1, 1)$, $(2, -1)$ has side vectors $(1, 1)$, $(1, -2)$, $(-2, 1)$, of squared lengths 2, 5, 5. The statement explicitly restricts to $p \equiv 1 \pmod{4}$, and we refute it as stated.